Applied Combinatorial Creativity

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1 Introduction

Combinatorial creativity (CC) is one of the three types of creativity described by Margaret Boden [3], alongside transformative and exploratory creativity. This is creativity that comes from unconventional combinations of two existing ideas. A graph-theoretical formulation for combinatorial creativity, used to evaluate LLM responses, was derived by Schapiro et al. [7].

The goal is to adapt the derivation to be useful for evaluating a Neuromorphic Multi-Agent Debate System (NeuMAD). This is a system of NeuKRAG agents [5]. This system has three NeuKRAG agents that each use a separate KG: one neuroscience, one neuromorphic, one AI/ML. They either debate and synthesize for N rounds, or generate responses and synthesize immediately in the case where $N = 0$.

The adapted metric needs to be such that a prompt optimizer such as GEPA [2] can be applied to improve upon this metric.

2 Combinatorial Creativity

Combinatorial creativity has two components:

- **Novelty**: how uncommon the combination between two ideas is.
- **Utility**: how applicable this combination is.

A creative artifact is a walk on a conceptual knowledge graph $G = (V, E, \mathcal{P})$, where V is the set of entities (nodes), E is the set of directed labeled edges (triples), and $\mathcal{P} : E \to [0, 1]$ assigns each triple a probability. The probability of a specific triple $e = (u, r, v)$ is written $p(e)$; it can be interpreted as a transition probability from u to v through relationship r.

A **walk** $P = (v_0, v_1, \ldots, v_h)$ is a sequence of h consecutive edges in E. The integer h is the walk length, and $\{\ell_1, \ldots, \ell_h\} = \{v_1, \ldots, v_h\}$ denotes the set of nodes visited (excluding the start node v_0).

A query given to an LLM has inclusionary constraints I and exclusionary constraints X, where $I, X \subset V$ and $I \cap X = \emptyset$.

Novelty is originally computed via the following formula:

$$N(P) := \alpha_h h + \alpha_r S(P)$$

where α_h and α_r are non-negative weights with $\alpha_h + \alpha_r = 1$, h is the walk length, and $S(P)$ is the surprise of the walk. Surprise is the average negative log-likelihood of the edges traversed:

$$S(P) = \frac{1}{h} \sum_{i=1}^{h} -\log\bigl(p(e_i)\bigr)$$

where e_i is the i-th edge of the walk.

Utility is originally computed via the following formula:

$$\begin{aligned} U(P; x) := {} & (1 + \alpha_I |I|)(1 + \alpha_X |X|) \\ & \cdot \mathbf{1}[v_0 = u,\ v_h = v,\ \{\ell_1, \ldots, \ell_h\} \supseteq I,\ \{\ell_1, \ldots, \ell_h\} \cap X = \emptyset] \end{aligned}$$

The indicator is 1 iff the walk visits every node in I (inclusionary constraint) and avoids every node in X (exclusionary constraint). Letting C be the boolean value of that indicator, the formula simplifies to:

$$U(P; x) := \begin{cases} (1 + \alpha_I |I|)(1 + \alpha_X |X|) & \text{if } C \\ 0 & \text{otherwise} \end{cases}$$

Combinatorial Creativity (CC) of a generative model $\mathcal{M}_\theta : \mathcal{T} \to \mathcal{H}$ (mapping prompts $\mathcal{T}$ to hypothesis responses $\mathcal{H}$), evaluated on a distribution $\mathcal{D}$ over prompts, is:

$$C(\theta) := \mathbb{E}_{x \sim \mathcal{D}}\bigl[U(\mathcal{M}_\theta(x); x) \cdot N(\mathcal{M}_\theta(x))\bigr]$$

For a single response P, this reduces to:

$$C(P) = U(P) \cdot N(P)$$

3 NeuMAD and NeuKRAG

NeuKRAG (Neuromorphic Knowledge-Retrieval-Augmented Generation) [5] is a closed-loop, KG-grounded hypothesis generation system for neuromorphic circuit design. Given a research query, an LLM-based entity extractor identifies key technical concepts, which are used as entry points for a breadth-first traversal of a domain knowledge graph. The retrieved subgraph—spanning spiking neuron models, learning rules, architectural motifs, and device technologies—is formatted as a set of (head, relation, tail) triples $T = \{(h_i, r_i, t_i)\}$ and injected into a DSPy `Predict` signature that synthesizes a grounded scientific hypothesis. Because every claim is anchored to explicit KG evidence, NeuKRAG avoids the hallucination and lack of grounding that plague unconstrained LLM generation. The system also supports a feedback loop in which simulation outcomes (via tools such as SuperNeuro and Fugu) are fed back to refine the KG and guide future hypothesis generation.

NeuMAD (Neuromorphic Multi-Agent Debate) extends NeuKRAG by instantiating three specialist agents—one each for neuroscience, AI/ML, and neuromorphic engineering—each grounded in its own domain KG. The system operates in two modes drawn from the Multi-Agent Debate (MAD) framework [6]: a *synthesis* mode in which agents produce independent hypotheses that a mediator integrates into a unified proposal, and an *adversarial* mode in which agents engage in tit-for-tat debate across up to N rounds before a judge extracts a final answer. The adversarial mode is motivated by the Degeneration-of-Thought (DoT) problem, where self-reflection causes an LLM to become overconfident in an incorrect answer; debate between agents with differing domain priors counteracts this collapse. A discriminative judge decides adaptively whether the debate has converged, and an extractive judge synthesizes the strongest evidence-grounded hypothesis from the full debate history.

For the purposes of evaluating combinatorial creativity, the walk P attributed to a NeuMAD response is operationalized as the *set* of KG nodes drawn from the fetched triples:

$$V(T) = \bigcup_i \{h_i,\, t_i\}$$

The ordering of these nodes into a coherent walk is handled by the procedure in Section 4.1, which also assigns transition probabilities to all pairs of nodes in $V(T)$.

4 Adapting CC

There are five issues when adapting this to the Neuromorphic Multi-Agent Debate system:

- The utility metric may or may not actually correlate with downstream utility.
- The utility metric is too harsh. If a single constraint isn't satisfied, the entire score goes to 0. A prompt optimizer will struggle to learn from such a strict signal.
- A RAG system is useful if it properly uses fetched information. This may need to be wired into the utility.
- The fetched triples may not form a contiguous structure. This will damage the walk length h used to compute novelty.
- The KG triples NeuKRAG uses don't have probabilities. Without this, we won't be able to compute surprise $S(P)$.

4.1 Assembling the Creative Landscape

Both novelty and utility involve the concept of a walk on a conceptual knowledge graph. However, the triples T fetched by NeuKRAG may not form a contiguous walk, and they carry no explicit edge probabilities—both of which are required by the original CC formulas. We resolve this by constructing a weighted graph over the fetched nodes and then finding a maximum-likelihood path through them.

Let $V = V(T) = \bigcup_i \{h_i, t_i\}$ be the set of nodes touched by the fetched triples. We assign a weight to every ordered pair (i, j) with $i, j \in V$:

$$w(i,j) = \begin{cases} 1 & \text{if } (i,j) \in E(G) \quad \text{(explicit KG edge)} \\ s(i,j) & \text{otherwise} \quad \text{(implicit linkage)} \end{cases}$$

For implicit linkages, $s(i, j)$ is the Adamic-Adar index [1], computed over the neighborhood structure of the full KG G:

$$s(i,j) = \sum_{u \in N(i) \cap N(j)} \frac{1}{\log |N(u)|}$$

where $N(i)$ is the set of neighbors of node i in G. Nodes that share many low-degree common neighbors receive a high score, reflecting that uncommon co-citations are a stronger signal of relatedness.

Let $\mathbf{W}$ be the $|V| \times |V|$ matrix with $\mathbf{W}_{ij} = w(i, j)$. Row-normalizing $\mathbf{W}$ gives the transition matrix $\mathbf{P}$:

$$\mathbf{P}_{ij} = \frac{\mathbf{W}_{ij}}{\sum_k \mathbf{W}_{ik}}$$

so that $\mathbf{P}_{ij} = p(i \to j)$ is the probability of stepping from node i to node j.

To connect all nodes into a single walk, we solve a minimum-weight perfect matching on the complete graph over V with edge costs $-\log \mathbf{P}_{ij}$ using the Blossom algorithm [4]. Minimizing total cost is equivalent to maximizing total log-probability, so the matching finds the *most likely* set of pairings through the fetched nodes. This is the correct objective here: we are not yet scoring novelty, we are assembling the most plausible walk that passes through the evidence the agent actually retrieved. Novelty is scored *afterwards* in Section 4.3, where a less likely walk earns a higher surprise $S(P)$. The matched pairs are chained sequentially into the path $\pi = (v_{\sigma(1)}, \ldots, v_{\sigma(|V|)})$, visiting every node in V exactly once.

4.2 Adapting Utility

There is a lingering question of whether measuring utility through constraint satisfaction even transfers well to utility through downstream application, i.e., "real utility". It may be that for the application of neuromorphic computing, a utility function might look something like this:

$$U^{\text{downstream}} = \frac{1}{2}\left(\mathbf{1}[\text{runs}] + \mathbf{1}[\text{learns}]\right) \cdot F$$

I.e., does the hypothesis yield a neuromorphic model that runs, learns, and adheres properly to the hypothesis itself, without needing to cut corners or diverge, as measured by a fidelity value F.

The issue with this is that it would be expensive and/or noisy to optimize via GEPA as it would require a downstream coding agent to run. In more expensive cases, we'd need people to implement generated hypotheses and have an alternative optimizer like PrefPO [8] to use human preference as signal.

For simplicity, we use a softened version of the original utility formulation. In this section, P denotes the *set* of KG nodes attributed to the response—concretely, $P = V(T)$ as defined in Section 3.

$$U(P) = \frac{|(I \cap P) \cup (X \setminus P)|}{|I \cup X|}(1 + \alpha_I|I|)(1 + \alpha_X|X|)$$

The numerator counts satisfied constraints: nodes in I that were visited ($I \cap P$) plus nodes in X that were successfully avoided ($X \setminus P$). All constraints met gives a full score; none met gives 0; intermediate cases are interpolated linearly.

To also reward proper use of retrieved evidence, we define a retrieval satisfaction term. Each paper $r \in R$ cited by the response has an associated KG subgraph $r_K \subseteq G$—the set of nodes and edges extracted from that paper during KG construction. The full utility score is:

$$\begin{aligned}
\tilde{U}(P) &= U_C(P) \cdot U_R(P) \\
U_C(P) &= \frac{|(I \cap P) \cup (X \setminus P)|}{|I \cup X|}(1 + \alpha_I|I|)(1 + \alpha_X|X|) \\
U_R(P) &= \frac{|\{r \in R \mid r_K \cap P \neq \emptyset\}|}{|R|}(1 + \alpha_R|R|)
\end{aligned}$$

$U_R(P)$ rewards the response for actually traversing nodes from each paper it cites: $r_K \cap P \neq \emptyset$ checks that at least one node from paper r's KG subgraph appears in the response walk.

4.3 Adapting Novelty

The two issues in computing novelty are that P may not be a single connected subgraph of the KG, and that the edges in P don't have explicit probabilities. A simple workaround is the average pairwise shortest-path distance over the $k = |P|$ distinct nodes in the walk:

$$N(P) = \frac{1}{\binom{k}{2}} \sum_{i<j} d(v_i, v_j)$$

where $d(v_i, v_j)$ is the shortest-path distance between nodes $v_i, v_j \in P$ in G. However, this is an entirely different metric which lacks grounding in the original literature and discards the structure of the surprise term.

Instead, using the transition matrix $\mathbf{P}$ and connected path π assembled in Section 4.1, the novelty score can be computed the same way as in the original Schapiro paper [7]:

$$N(P) = \alpha_h h + \alpha_r S(P)$$

where $h = |\pi| - 1$ is the length of the assembled path and $S(P)$ is evaluated using $\mathbf{P}_{ij}$ as the edge transition probabilities along π.

5 Synthesis

The derivations above yield four algorithms: assembling the creative landscape from fetched triples, computing utility, computing novelty, and combining them into a final CC score.

Algorithm 1 Assemble Creative Landscape

Require: fetched triples $T = \{(h_i, r_i, t_i)\}$, full KG G
Ensure: transition matrix $\mathbf{P}$, connected path π

$V \leftarrow \bigcup_i \{h_i, t_i\}$ ▷ nodes touched by fetched triples
for each ordered pair (i, j) with $i, j \in V,\ i \neq j$ **do**
  if $(i, j) \in E(G)$ **then**
    $\mathbf{W}_{ij} \leftarrow 1$
  else
    $\mathbf{W}_{ij} \leftarrow \sum_{u \in N(i) \cap N(j)} \frac{1}{\log |N(u)|}$ ▷ Adamic-Adar over G
  end if
end for
$\mathbf{P}_{ij} \leftarrow \mathbf{W}_{ij} \,/\, \sum_k \mathbf{W}_{ik}$ ▷ row-normalise to get transition probabilities
$M \leftarrow \text{Blossom}(V,\ \text{cost}(i,j) = -\log \mathbf{P}_{ij})$ ▷ min-cost = max-likelihood matching over fetched nodes
$\pi \leftarrow \text{ChainMatching}(M)$ ▷ concatenate matched pairs into a single walk over V
return $\mathbf{P}$, π

References

[1] Lada A Adamic and Eytan Adar. "Friends and neighbors on the web". en. In: *Soc. Networks* 25.3 (July 2003), pp. 211–230.

[2] Lakshya A Agrawal et al. *GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning.* 2026. arXiv: `2507.19457 [cs.CL]`. URL: `https://arxiv.org/abs/2507.19457`.

Algorithm 2 Compute Utility

Require: node set $P = V(T)$, inclusionary constraints I, exclusionary constraints X, paper-to-subgraph map $\{r_K\}$, cited papers R, weights $\alpha_I, \alpha_X, \alpha_R$

Ensure: utility score $\tilde{U}$

$\text{sat} \leftarrow |(I \cap P) \cup (X \setminus P)| \;/\; |I \cup X|$ ▷ fraction of constraints satisfied

$U_C \leftarrow \text{sat} \cdot (1 + \alpha_I |I|)(1 + \alpha_X |X|)$

$\text{used} \leftarrow |\{r \in R \mid r_K \cap P \neq \emptyset\}|$ ▷ papers whose KG subgraph was traversed

$U_R \leftarrow (\text{used} \;/\; |R|) \cdot (1 + \alpha_R |R|)$

return $\tilde{U} \leftarrow U_C \cdot U_R$

Algorithm 3 Compute Novelty

Require: connected path $\pi = (v_0, v_1, \ldots, v_h)$, transition matrix $\mathbf{P}$, weights α_h, α_r

Ensure: novelty score N

$S \leftarrow \dfrac{1}{h} \displaystyle\sum_{i=0}^{h-1} -\log \mathbf{P}_{v_i,\, v_{i+1}}$ ▷ average surprise along π

return $N \leftarrow \alpha_h \cdot h + \alpha_r \cdot S$

Algorithm 4 Compute Combinatorial Creativity

Require: fetched triples T, full KG G, constraints I, X, cited papers R, paper-to-subgraph map $\{r_K\}$, weights $\alpha_I, \alpha_X, \alpha_R, \alpha_h, \alpha_r$

Ensure: CC score C

$\mathbf{P},\ \pi \leftarrow \textsc{AssembleCreativeLandscape}(T,\ G)$

$\tilde{U} \leftarrow \textsc{ComputeUtility}(V(T),\ I,\ X,\ \{r_K\},\ R,\ \alpha_I,\ \alpha_X,\ \alpha_R)$

$N \leftarrow \textsc{ComputeNovelty}(\pi,\ \mathbf{P},\ \alpha_h,\ \alpha_r)$

return $C \leftarrow \tilde{U} \cdot N$

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